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 RV UNIVERSITY <i>Go, change the world</i> an initiative of RV EDUCATIONAL INSTITUTIONS	School of Computer Science and Engineering Minor in Robotics and Industrial Automation CIE - 3: Project Academic Year 2025 – 2026	
	Course: Perception & Localization	Course Code: CS3112
Date: 13 Nov. 2025	Duration:	Semester: V Max Marks: 20 Marks

Project:

Implementation of Camera-Based SLAM using Extended Kalman Filter (EKF) in Python.

Problem Statement:

- In robotic perception and navigation, Simultaneous Localization and Mapping (SLAM) allows a robot to build a map of an unknown environment while simultaneously estimating its own position within that map.
- This project focuses on implementing a camera-based EKF-SLAM system, where a robot equipped with a monocular or stereo camera must estimate both its pose (position and orientation) and the landmark locations in the environment.
- Students will implement the core components of EKF-SLAM using Python and visualize the robot's motion, landmark observations, and map construction in Pygame (or an equivalent simulation environment).

S.No.	Statement	Marks	Level	CO
1	Tasks: a) Model the Robot Motion: (a) Define a nonlinear motion model for a differential-drive robot. Given the robot's state: $\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$ and control inputs : linear velocity, v_c and angular velocity, ω_c, the motion model (nonlinear) for the differential drive robot is given as: $\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{bmatrix} = \begin{bmatrix} x_t + v_c \Delta t \cos \theta \\ y_t + v_c \Delta t \sin \theta \\ \theta_t + \omega_c \Delta t \end{bmatrix} + \epsilon_t$ Where $\epsilon(t)$ is noise. Each landmark i at position (x_i, y_i) is observed at range r_i and bearing angle, ϕ_i (relative to robots heading) : $\begin{bmatrix} r_i \\ \phi_i \end{bmatrix} = \begin{bmatrix} \sqrt{(x - x_i)^2 + (y - y_i)^2} \\ \arctan\left(\frac{y - y_i}{x - x_i}\right) - \theta \end{bmatrix} + \delta_t$ If one assumes pure camera, then ranging will not be there and only bearing will remain. (b) Simulate movement based on velocity and angular velocity commands. b) Simulate the Environment: (a) Create a 2D world with fixed visual landmarks (points). Take 3 to 4 points. (b) Simulate noisy camera observations of landmarks (e.g., bearing or pixel position). c) Implement EKF-SLAM: (a) Prediction step: Propagate the robot's state and covariance. (b) Update step: Use camera observations to correct the estimates. (c) Maintain a combined state vector for both robot and landmarks. d) Visualize in Pygame: (a) Display the robot (true and estimated positions), landmarks (true and estimated), and uncertainty ellipses. (b) Show how the map improves as the robot moves.	1	L6	CO4

S.No.	Statement	Marks	Level	CO
1	Prepare a Project Report: Template enclosed in GCR. Highlight the following: a) Problem description and EKF equations. b) Implementation steps and results. Screenshots of code and Pygame. c) Plots of estimated vs. true trajectories and landmarks. d) Discussion on limitations and possible improvements. Marking Scheme: 1 EKF Implementation: Correct use of nonlinear models, Jacobians, and update logic - 5Marks. 2 Simulation and Visualization: Clear display of robot and map in Pygame - 5Marks 3 Results and Analysis: Accuracy of estimation and discussion of uncertainty. - 5Marks 4 Report: Clarity and completeness of documentation. - 5Marks 5 (Bonus): Creativity/ Extensions: Added realism (e.g., field of view, data association, camera noise model). - 5Marks		L6	CO4